

Exercise 8

Zhuge X.K.

January 7, 2026

Exercise 3.4

Exercise 3.6

Derive

$$\hat{G}^R(\tau) = -\frac{i}{\hbar} \theta(\tau) e^{-\frac{i}{\hbar} E_n \tau} |\psi_n\rangle \langle \psi_n|$$

using the following expansion

$$\hat{G}^R(E) = \frac{1}{(E + i\eta)\hat{I} - \hat{H}} \sum_n |\psi_n\rangle \langle \psi_n| = \frac{1}{E - E_n + i\eta} \sum_n |\psi_n\rangle \langle \psi_n|$$

Exercise 3.7

For 1D free electron

$$\left[E \pm i\eta + \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \right] G^{R/A}(x, x'; E) = \delta(x - x')$$

Find two exact solution: $G_1(x - x', E) = G^R$ corresponding to "+", and $G_2(x - x', E) = G^A$ corresponding to "-".